

Shriman Oppilamani

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EDUCATION

Arizona State University

May 2026

Bachelor of Science in Computer Science, GPA 4.10/4

SKILLS

Languages: Rust, C++, Java, Python, HTML, CSS, Javascript, SQL, C#, GDScript

Technologies: MySQL, Unity3D, Linux, React, TailwindCSS, Unreal Engine, FastAPI, Godot, MariaDB

PROJECTS

Melody OS (Custom Embedded Audio System)

Jan 2026 - Present

Linux, Python, Systemd

- Designed and deployed a lightweight, headless operating system on Raspberry Pi dedicated to high-fidelity audio playback.
- Engineered a custom service layer using Mopidy and Iris, exposing a local web interface for network-wide queue management and lossless playback from local HDD/YouTube.
- Configured low-latency audio pathways (ALSA) and managed headless boot processes to ensure "plug-and-play" functionality without peripherals.

Low-Level Text Editor

Nov 2025 - Present

C, Make, Unix Systems

- Building a text editor from scratch in C, directly manipulating termios structures to handle raw input modes and disable canonical processing.
- Implementing a custom TUI (Text User Interface) using ANSI escape sequences to handle cursor positioning, screen refreshing, and scrolling logic.
- Manually managing dynamic memory allocation for row buffers and file I/O operations, ensuring memory safety without high-level abstractions.

Haven (Full-Stack Wellness Platform)

Sept 2025

React, Gemini API, Tailwind

- Developed a responsive web app integrating the Gemini API to analyze journal entries and generate empathetic, context-aware venting feedback.
- Implemented local storage strategies to ensure user data privacy while delivering personalized meditation prompts and mood tracking.

Custom Pseudocode Compiler – Solo Class Project

April 2025 - May 2025

C++

- Built a compiler in C++ to parse and execute pseudocode with loops, conditionals, switches, and arithmetic operations.
- Designed a comprehensive Abstract Syntax Tree (AST) and context-free grammar before implementation to ensure modular code generation.
- Implemented components for lexical scanning, parsing, and execution, applying compiler theory in practice.
- Demonstrated ability to independently design and deliver complex systems' software.

Linux Kernel and Hardware Optimization

Dec 2024 - Present

Arch Linux, Bash

- Migrated Microsoft Surface hardware to a custom Arch Linux environment, patching the kernel to resolve critical thermal throttling (CPU locking at 200MHz).
- Diagnosed and resolved WPA2-Enterprise (Eduroam) authentication failures by debugging wpa-supPLICANT driver capabilities.
- Optimized power management profiles and solved proprietary driver conflicts to ensure stable daily operation on unsupported hardware.